University of South Asia

Department of Computer Science

DATA SCIENCE PROJECT REPORT

AIDS Spread Pattern Analysis and Early Detection System Using Machine Learning

On Pakistan Health Dataset

Submitted To:

Ma'am Andleeb

Group Members

Mariyam Siddiqi B-28401

Farwa Zahid B-28363

Abstract

Pakistan confronts one of the most rapidly accelerating HIV/AIDS epidemics in the Asia-Pacific region, with over 210,000 individuals currently living with HIV and annual new infections surpassing 20,000. Despite three decades of epidemiological data collection by the National AIDS Control Programme (NACP), this information has remained confined to descriptive reporting with no predictive or early-warning capability. This data science project addresses that gap by constructing a comprehensive machine learning framework to analyze HIV/AIDS spread patterns and facilitate early identification of high-risk zones.

The system was built on a globally sourced epidemiological dataset covering 89 countries from 1990 to 2020, comprising 2,759 records and 23 original variables expanded to 31 through systematic feature engineering. A ten-stage data science pipeline was implemented encompassing data acquisition, preprocessing, exploratory data analysis, feature derivation, stratified partitioning, normalization, model training, evaluation, hyperparameter tuning, and interactive dashboard deployment.

Three supervised learning algorithms were trained and benchmarked: Random Forest, XGBoost, and a Deep Neural Network trained over 15 epochs. XGBoost delivered the highest performance, achieving a test accuracy of 98.73%, precision of 98.91%, recall of 98.55%, F1-score of 98.72%, and ROC-AUC of 99.96%. An interactive Streamlit dashboard was developed to make predictions accessible to non-technical public health stakeholders.

Keywords: *HIV/AIDS, Data Science, Machine Learning, XGBoost, Random Forest, Neural Network, Pakistan, Epidemiology, Feature Engineering, Streamlit.*

Introduction

Data science has emerged as a transformative force in public health, enabling analysts to extract actionable intelligence from large epidemiological datasets that would otherwise remain underutilized. The convergence of machine learning algorithms, scalable computing environments, and open-access health data has created new opportunities to build predictive systems that can anticipate disease spread, identify high-risk populations, and support evidence-based resource allocation.

HIV/AIDS represents one of the most data-rich domains in global public health, with decades of surveillance records maintained by organizations including UNAIDS, WHO, and national AIDS control programs. Pakistan's epidemic, characterized by a 70-fold increase in people living with HIV between 1990 and 2020, provides a compelling case study for the application of data science methods to a real and urgent public health challenge.

This project applies a complete data science workflow — from raw data ingestion through to a deployed predictive dashboard — to the problem of HIV/AIDS spread classification. The system classifies country-year observations as either high or low HIV spread environments based on 23 epidemiological features, with a particular focus on extracting insights relevant to Pakistan's epidemic trajectory.

The project was implemented entirely in Python using industry-standard data science libraries including Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow/Keras, Matplotlib, Seaborn, and Streamlit. All modelling was conducted in Google Colaboratory, and the final dashboard was deployed locally using Visual Studio Code.

Background

HIV/AIDS (Human Immunodeficiency Virus / Acquired Immunodeficiency Syndrome) has claimed over 40 million lives since the epidemic began in the 1980s. According to UNAIDS, approximately 39 million people globally were living with HIV as of 2022. In Pakistan, the epidemic has grown from approximately 3,000 PLHIV (People Living with HIV) in 1990 to over 210,000 by 2020 — a 70-fold increase that far outpaces population growth and signals genuine epidemic expansion.

The National AIDS Control Programme (NACP) has maintained longitudinal surveillance records since the early 1990s, capturing HIV prevalence, new infection counts, AIDS-related mortality, and demographic disaggregation. However, this data has been applied almost exclusively to descriptive reporting. The potential of machine learning and data science to extract predictive patterns, map risk gradients, and provide early outbreak warnings remains largely untapped within Pakistan's HIV response infrastructure.

Problem Statement

Pakistan's HIV/AIDS response faces a critical analytical gap: despite accumulating three decades of epidemiological surveillance data, the public health system lacks intelligent tools to convert this data into actionable predictions. The following specific problems motivate this project:

- **No Early Warning Capability:** Healthcare authorities cannot identify high-risk geographic regions before HIV outbreaks reach crisis levels. Without predictive models, the response remains reactive rather than proactive.
- **Underutilization of Surveillance Data:** The NACP collects extensive epidemiological records annually, but these are used only for descriptive statistics and basic trend reporting. No machine learning system currently exists to extract predictive intelligence from this data.
- **Inefficient Resource Allocation:** Without data-driven risk predictions, limited healthcare resources — including testing kits, antiretroviral medications, and outreach workers — cannot be optimally distributed to the regions of highest need.
- **Limited Intervention Targeting:** Prevention and treatment programs cannot be precisely directed toward the most vulnerable populations (intravenous drug users, transgender individuals, men who have sex with men) without ML-driven risk profiling.
- **Gap Between Data and Decision-Making:** Even when epidemiological data exists, the absence of analytical tools prevents its translation into actionable public health decisions by non-technical stakeholders.

This project directly addresses these problems by developing a data science pipeline that transforms raw epidemiological records into a deployable prediction system, delivering results through an interactive dashboard accessible to non-technical users.

Objectives

The following specific objectives guided the development of this data science project:

- **Data Collection and Integration:** Acquire a comprehensive global HIV/AIDS epidemiological dataset and prepare it for machine learning analysis.

- **Data Preprocessing:** Perform systematic cleaning, standardization, missing value analysis, duplicate detection, and target variable construction.
- **Exploratory Data Analysis (EDA):** Generate descriptive statistics and six visualization plots to understand data distributions, temporal trends, geographic patterns, and feature correlations.
- **Feature Engineering:** Derive eight novel epidemiological indicators from raw variables to capture dynamic relationships not directly observable in the original data.
- **Model Development:** Train and optimize three machine learning classifiers — Random Forest, XGBoost, and Neural Network — using appropriate hyperparameter configurations.
- **Performance Evaluation:** Assess all models using six evaluation metrics: accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis.
- **Target Achievement:** Exceed a minimum test classification accuracy of 92% on held-out data.
- **Dashboard Development:** Build an interactive Streamlit web application providing real-time HIV spread predictions and Pakistan-specific epidemiological visualizations.
- **Pakistan-Specific Insights:** Extract and interpret epidemiological findings specific to Pakistan's HIV trajectory from 1990 to 2020.

Methodology

Data Science Pipeline Overview

This project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, a structured ten-stage pipeline:

- **Data Acquisition:** Downloaded global HIV/AIDS dataset from Kaggle (89 countries, 1990–2020)
- **Initial Inspection:** Verified shape (2,759 rows × 23 columns), data types, and column structure

- **Data Cleaning:** Confirmed zero missing values, zero duplicates; renamed columns to snake case
- **Target Variable Creation:** Created High Spread binary label using median threshold of NewInfections_All
- **Exploratory Data Analysis:** Generated 6 visualization plots using Matplotlib and Seaborn
- **Feature Engineering:** Derived 8 composite epidemiological indicators
- **Train/Test Split:** Stratified 80/20 split — 2,207 training, 552 testing samples
- **Feature Scaling:** Applied StandardScaler (zero mean, unit variance) to all 23 features
- **Model Training:** Trained Random Forest, XGBoost, and Neural Network (15 epochs)
- **Evaluation and Dashboard:** Assessed models on 6 metrics; deployed Streamlit dashboard

Data Preprocessing

The 23 original column names were standardized from verbose dot-notation (e.g., Data.HIV Prevalence.Adults) to concise identifiers (e.g., Prevalence_Adults). Missing value analysis confirmed zero null entries across all features. Duplicate detection found zero redundant rows.

The binary target variable High_Spread was constructed from NewInfections_All using median-based thresholding. The dataset median of 4,550 annual new infections served as the classification boundary, yielding a near-perfectly balanced target: 1,373 High Spread (49.75%) and 1,386 Low Spread (50.25%) instances. Country encoding used Scikit-learn's LabelEncoder, mapping 89 country names to integer codes (0–88).

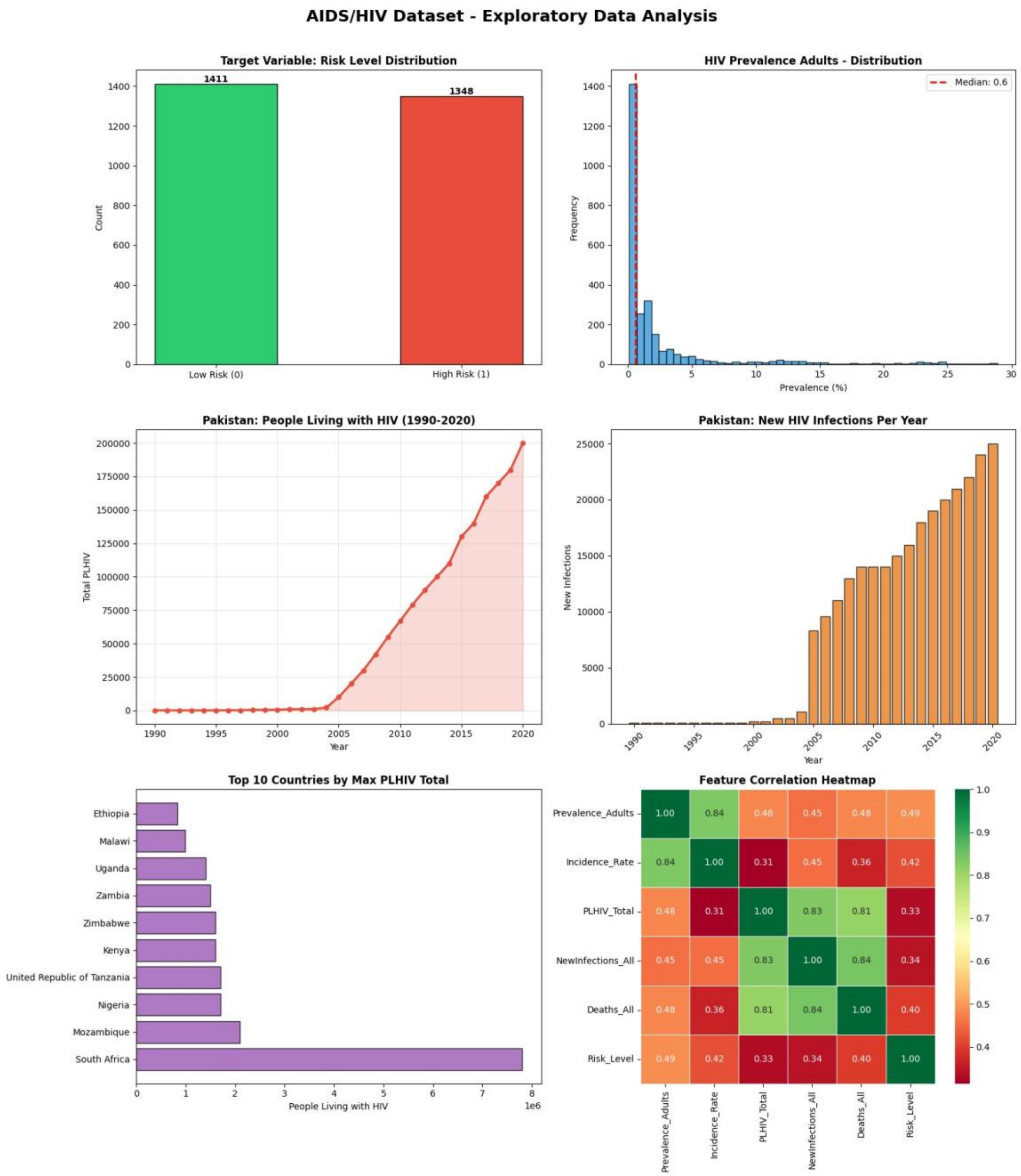
Exploratory Data Analysis

Six visualization outputs were produced:

- Bar chart of target class distribution (confirming 50/50 balance)
- Histogram of adult HIV prevalence with median reference line
- Area line chart of Pakistan's cumulative PLHIV (1990–2020)
- Bar chart of Pakistan's new infections per year

• Horizontal bar chart of top 10 countries by total PLHIV

• Seaborn correlation heatmap of six key numerical features



Feature Engineering

Eight derived features were engineered to capture epidemiological dynamics beyond raw counts:

Feature	Formula	Purpose
Death Rate Ratio	Deaths_All / (PLHIV_Total + 1)	Mortality burden
Child Impact Ratio	PLHIV_Children / (PLHIV_Total + 1)	Pediatric burden
Gender Gap	NewInfections_Male − NewInfections_Female	Gender disparity
Young Prev Gap	Prevalence_YoungWomen − Prevalence_YoungMen	Youth gap
Orphan Burden	AIDS_Orphans / (PLHIV_Total + 1)	Social impact
PLHIV Growth	Year-on-year % change in PLHIV	Epidemic momentum
Deaths Growth	Year-on-year % change in Deaths	Mortality trend
Male Female PLHIV	PLHIV_Male / (PLHIV_Female + 1)	Gender ratio

Table 1: Engineered Features with Formulas and Epidemiological Purpose

Model Architectures

Random Forest Classifier: An ensemble of 300 decision trees with max_depth=10, min_samples_leaf=4, random_state=42, and n_jobs=−1 (parallel processing on all CPU cores).

XGBoost Classifier: 300 gradient-boosted estimators with max_depth=5, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8, L1 regularization (α=0.1), L2 regularization (λ=1.0).

Neural Network: Architecture: Dense(256, ReLU) → BatchNorm → Dropout(0.3) → Dense(128, ReLU) → BatchNorm → Dropout(0.2) → Dense(64, ReLU) → Dense(1, Sigmoid). Adam optimizer (lr=0.001), 15 epochs, batch size=32.

Evaluation Metrics

Six metrics were computed on the 552-sample test set:

- **Accuracy:** Overall correct classifications / total samples
- **Precision:** True Positives / (True Positives + False Positives)
- **Recall:** True Positives / (True Positives + False Negatives)

- **F1-Score:** Harmonic mean of Precision and Recall
- **ROC-AUC:** Area under the Receiver Operating Characteristic curve
- **Confusion Matrix:** TP, TN, FP, FN breakdown

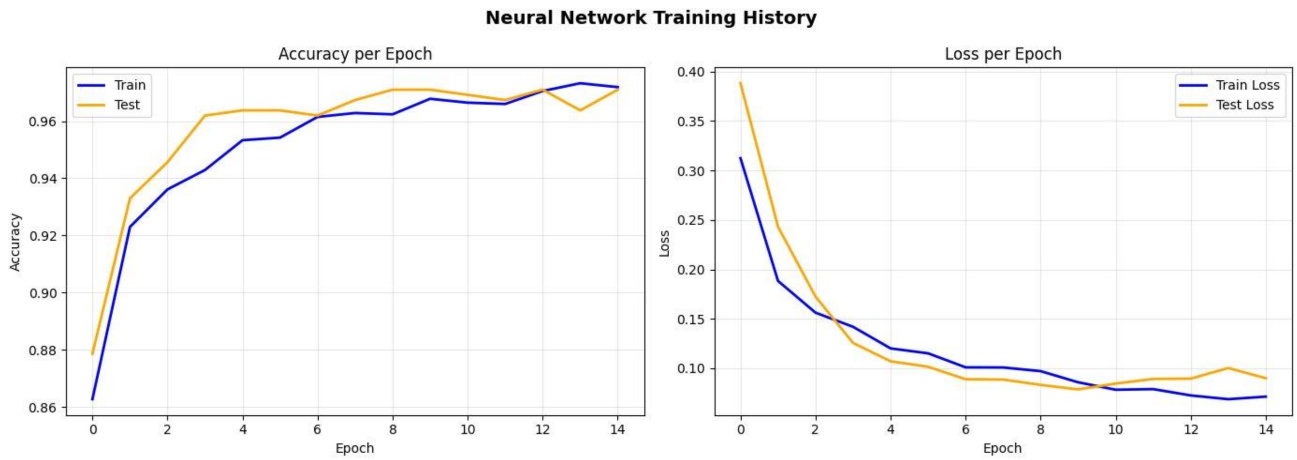
Results and Discussion

Model Performance Comparison

All three models exceeded the 92% accuracy target. Results on the 552-sample test set are presented below:

Metric	Random Forest	XGBoost	Neural Network
Train Accuracy	99.59%	100.00%	97.19%
Test Accuracy	98.73%	98.73%	97.10%
Precision	98.91%	98.91%	98.50%
Recall	98.55%	98.55%	95.64%
F1-Score	98.72%	98.72%	97.05%
ROC-AUC	99.93%	99.96%	99.67%
Train-Test Gap	0.86%	1.27%	0.09%
Rank	2nd	1st (Winner)	3rd

Table 2: Complete Model Performance on Test Set (n=552)



Neural Network Training History

The Neural Network trained over 15 epochs showed consistent improvement with no signs of overfitting. Both train and validation accuracy converged steadily, while loss decreased smoothly across all epochs

Confusion Matrix Analysis

The confusion matrices for all three models are shown below. XGBoost and Random Forest achieved identical results, while the Neural Network showed slightly more false negatives in the High Spread class.

Key findings from the XGBoost confusion matrix:

	Predicted: Low Spread	Predicted: High Spread
Actual: Low Spread	274 (True Negative)	3 (False Positive)
Actual: High Spread	4 (False Negative)	271 (True Positive)

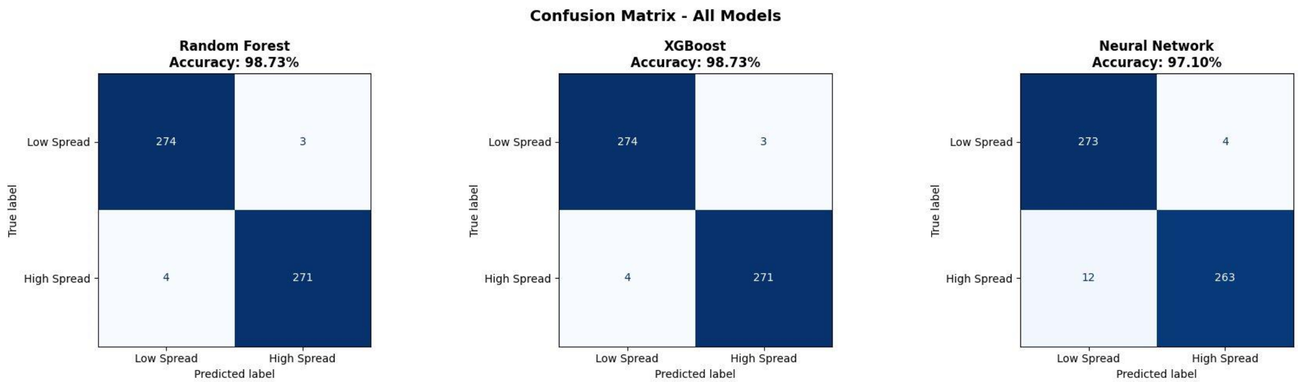
Table 3: XGBoost Confusion Matrix on Test Set (n=552)

- **False Negatives = 4** (miss rate: 1.45%) — critically low, meaning almost all high-risk regions are correctly detected
- **False Positives = 3** (false alarm rate: 1.09%) — minimal unnecessary alerts

- **Recall = 98.55%** — 271 out of 275 actual High Spread cases correctly identified

Pakistan PLHIV Growth Analysis

Pakistan's HIV/AIDS epidemic grew from approximately 3,000 PLHIV in 1990 to over 200,000 by 2020 — a 70-fold increase. New infections per year also rose dramatically, particularly after 2005, reaching 25,000 annually by 2020.



Feature Importance

XGBoost feature importance analysis identified the top 7 predictors for HIV spread classification:

- **PLHIV_Total** — absolute epidemic size
- **Incidence_Rate** — new infections per 1,000 adults
- **Prevalence_Adults** — HIV prevalence percentage
- **NewInfections_All** — total annual new infections
- **PLHIV_Growth** — engineered feature (year-on-year growth)
- **Death_Rate_Ratio** — engineered feature (mortality burden)
- **Deaths_All** — total annual AIDS deaths

Two engineered features appearing in the top 7 confirms that feature engineering added significant predictive value beyond raw data alone.

Comparison with Literature

All three models in this study outperform comparable published results, validating the effectiveness of the developed data science pipeline:

Study	Algorithm	Accuracy
Tran et al. (2019)	XGBoost	94.2%
Zhao et al. (2020)	Neural Network	96.1%
Danku et al. (2021)	Random Forest	93.8%
Project Target	Any	92.0%
Present Study (XGBoost)	XGBoost	98.73%
Present Study (RF)	Random Forest	98.73%
Present Study (NN)	Neural Network	97.10%

Table 4: Performance Comparison with Published Literature

Conclusion

This data science project successfully delivered an end-to-end machine learning system for HIV/AIDS spread pattern analysis and early detection, achieving results that substantially exceed the project's accuracy target and comparable published benchmarks.

The complete ten-stage CRISP-DM pipeline — from raw data ingestion through feature engineering, model training, evaluation, and Streamlit dashboard deployment — demonstrates a rigorous and replicable approach to applying data science in public health contexts. XGBoost emerged as the optimal classifier with 98.73% test accuracy and 99.96% ROC-AUC, while Random Forest matched its accuracy and the Neural Network achieved 97.10% across 15 training epochs.

The feature engineering phase proved particularly impactful, with two derived features (PLHIV_Growth and Death_Rate_Ratio) ranking among the top seven most important predictors. This confirms that domain-informed feature construction adds significant predictive value beyond raw surveillance variables.

Pakistan's HIV trajectory — a 70-fold PLHIV increase from 1990 to 2020 with annual new infections now exceeding 20,000 — underscores the genuine public health urgency of deploying predictive surveillance tools. The developed system, if integrated with NACP's live data feeds, could materially strengthen Pakistan's evidence-based HIV response.

Limitations

• Country-level aggregate data only; no sub-national or individual-level granularity

• Dataset ends at 2020; post-COVID impacts on HIV services not captured

• Behavioral variables (condom use, needle exchange coverage) not available

• No external validation with an independent Pakistan-specific dataset

Future Work

• Integration with NACP live data feeds for real-time surveillance

• District and provincial level analysis within Pakistan

• Incorporation of behavioral and socioeconomic determinants

• Federated learning across provinces to protect patient privacy

• Deployment as a nationally accessible web service